

Lab 5: State Diagram

Online Railway Ticket Reservation System

A national online railway reservation system allows passengers to search for trains and select seats. Once a seat is selected, the system temporarily blocks the seat and starts a countdown timer during which the passenger must complete payment. During this time, internal activities such as fare calculation and seat availability monitoring continue.

Passenger validation and payment processing occur concurrently. If payment succeeds within the allowed time, the ticket is confirmed and issued. If payment fails or the timer expires, the seat is released automatically. After confirmation, the passenger may cancel the ticket, triggering refund processing and seat release.